# SOWING THE SEEDS OF CLIMATE RESILIENT INDIGENOUS FOOD SYSTEMS IN NORTHWESTERN ONTARIO

## ADDRESSING BARRIERS TO FOOD SECURITY AND ENVISIONING PATHWAYS FORWARD

NETMIZAAGGAMIG
NISHNAABEG
RED ROCK
INDIAN BAND
Strength. Tradition. Empowerment.
BIINJITIWAABIK
ZAAGING
ANISHINAABEK

# INTRODUCTION

In Northern Ontario, Indigenous communities are exposed to complex challenges that shape the availability, accessibility, and affordability of food.

Climate change threatens to add greater complexity by driving up food prices, changing ecosystem dynamics, and making environmental conditions less predictable.

# CONTEXT

## What are the unique stressors that impact Indigenous food systems?

Nutritional Colonialism (Mosby & Galloway, 2017).

Poverty and Extractivism (Ferguson et al., 2022)

Insufficient Approaches to Food Insecurity (Tarasuk et al., 2020)

# CONTEXT

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# RESEARCH QUESTIONS

**What are some key strengths, weaknesses, and threats in each community's food system?**

**How might climate change interact with them?**

# METHODOLOGY AND METHODS

| Methodology | Data Collection | Data Analysis |
|---|---|---|
| **Relationality.**<br><br>**Systems Thinking.**<br><br>**One Health and Anishinaabe Perspectives of Health.**<br><br>**Participatory Action Research.** | • Semi-structured **focus group** discussions<br>• **Interviews** with community leaders and garden stewards<br>• **Participatory mapping** of the food system | • Reflexive **thematic analysis** |
| | • Keyword search of the **Ontario Hansard** database (2018-2025) to isolate relevant **debates**.<br>• Keyword search in **Google Scholar** to isolate relevant **journal articles**. | • **Content analysis** of Hansard Transcripts<br>• **Narrative review** of journal articles about the impact of climate change on traditional foods |

**CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS**

**THREATS TO FOOD SECURITY**

**ENVISIONING PATHWAYS FORWARD**

# CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS

## What are some key strengths, weaknesses, and threats in each community's food system?

"I know we have to start teaching our young kids... That's the only way we'll keep our traditions, by teaching the young ones. If we're all gone, they won't know how to forage or what to forage." -RRIB

"It wasn't too long ago that I started learning about my ways because my mom's teachings about being a native person weren't too good. I used to hold that against her, but now I know it wasn't her fault. She was brainwashed to think that her culture wasn't good." -NN

# CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS

## What are some key strengths, weaknesses, and threats in each community's food system?

"We've always had a problem because we're semi-isolated. We have stores in Nipigon, but the prices are really high. A lot of people prefer going to Thunder Bay because they have better stores and better-quality food. However, many people don't have vehicles, so it's hard for them. " -BZA

# CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS

## What are some key strengths, weaknesses, and threats in each community's food system?

"If it's not from Lake Nipigon, I don't eat it" –BZA

# CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS

## What are some key strengths, weaknesses, and threats in each community's food system?

"You used to catch really good fish there, but they say you don't anymore because they haven't had the chance to grow that big." –NN

# RESEARCH QUESTIONS

**What are some key strengths, weaknesses, and threats in each community's food system?**

**How might climate change interact with them?**

Hansard Transcript Analysis
Thematic Analysis of Focus Group, Interview, and Participatory Mapping Data
Literature Review

# THREATS TO FOOD SECURITY

## CLIMATE CHANGE AND TRADITIONAL FOODS

- Changes in aquatic environments may be detrimental to the health of cool- and cold-water fish (Collingsworth et al., 2017).
- Climate change will disrupt food chains. There may be competition between newly encroaching species and existing ones (Reich et al., 2022).
- Changes in seasonality may occur, and disturbances may become more frequent (Bergeron et al., 2023).
- Changing precipitation patterns may occur, and wetland environments may become drier (Chu, 2015; Derksen et al., 2019).

# THREATS TO FOOD SECURITY

## CLIMATE CHANGE AND STORE-BOUGHT FOODS

- Agricultural productivity is expected to decrease across much of the globe's arable land (Dang et al., 2025; Luchtenbelt et al., 2025; Mirón et al., 2023).

- Insufficient nutrients are available to ensure global food security, even if food were perfectly distributed (Holt-Giménez et al., 2012; KC et al., 2018; Latham, 2021; Tomlinson, 2013).

- The value of farm, fishing, and intermediate food product imports to Canada increased by 11.6% in 2024, primarily driven by higher prices for nuts, vegetables, fresh fruits, and pulses (Global Affairs Canada, 2025).

# HOW CAN FOOD SYSTEM RESILIENCE BE BUILT IN NETMIZAAGGAMIG NISHNAABEG, BIINJITIWAABIK ZAAGING ANISHINAABEK FIRST NATION, AND RED ROCK INDIAN BAND AS CLIMATE CHANGE PROGRESSES?

# ENVISIONING PATHWAYS FORWARD

Hansard Transcript Analysis

Thematic Analysis of Focus Group, Interview, and Participatory Mapping Data

Literature Review

# ENVISIONING PATHWAYS FORWARD

Hansard Transcript Analysis
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# THANK YOU!

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UNIVERSITY of GUELPH
SCHOOL OF ENVIRONMENTAL DESIGN AND RURAL DEVELOPMENT

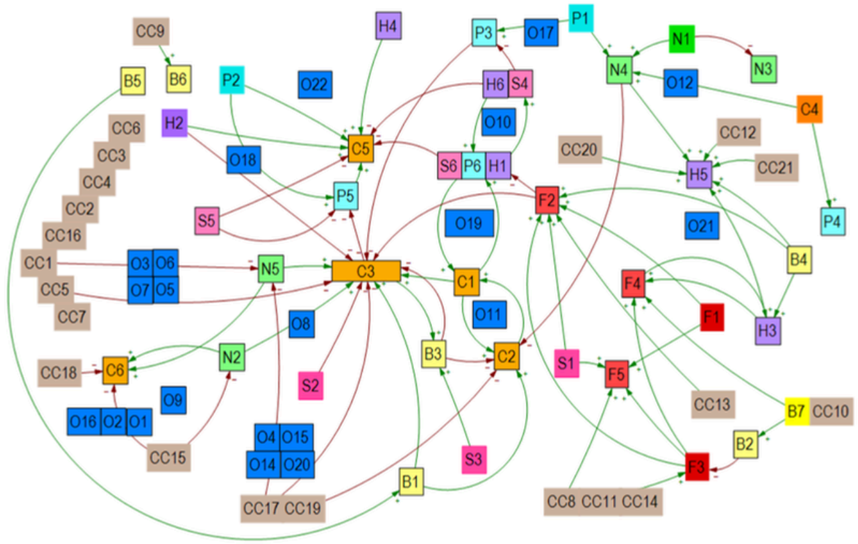
Ontario Agri-Food Innovation Alliance

NSERC
RESNET

Sioux Lookout
Biinjitiwaabik Zaaging Anishinaabek
Franz's 250 Main, Beardmore, ON
Red Rock Indian Band
Hearst
Zechner's Food Market, Nipigon, ON
Your Independent Grocer, Manitouwadge, ON
Kapuskasing
Your Independent Grocer, Marathon, ON
Cochrane
Real Canadian Superstore, Thunder Bay, ON
Netmizaaggamig Nishnaabeg
North Central Co-Op, White River, ON
Thunder Bay
Timmins
Your Independent Grocer, Wawa, ON
Lake Superior
Houghton
Pino's Get Fresh, Sault Ste. Marie, ON
Marquette
Sudbury

# Systems Thinking.

# Relationality.




# One Health and Anishinaabe Perspectives of Health.

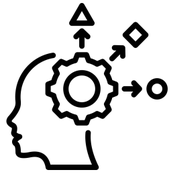
# Participatory Action Research.

# ENVISIONING PATHWAYS FORWARD

## What makes a food system resilient?

**Robustness**

**Recovery**

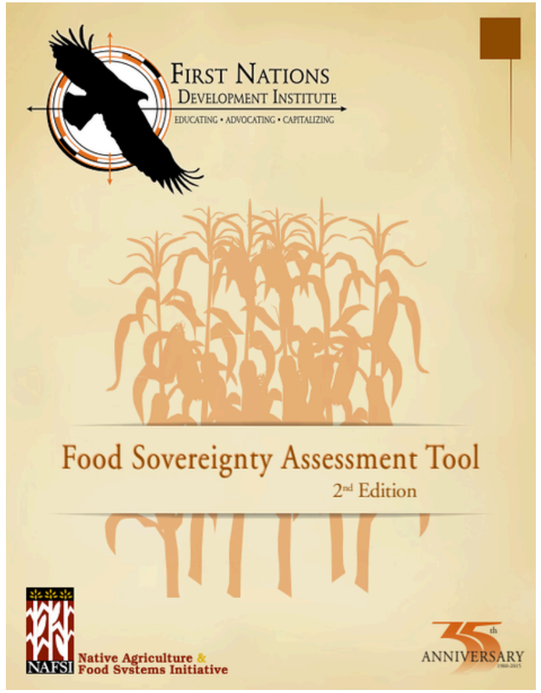


**Reorientation**

(HLPE-FSN, 2025; Zurek et al., 2022).

# CHARACTERIZATION OF THE COMMUNITY FOOD SYSTEMS

## What are some key strengths, weaknesses, and threats in each community's food system?




**Mapping Workshop and Focus Group Discussion Procedure**

Materials: Two community maps printed per community (small scale and large scale), markers, sticky notes, and push pins. Consent forms.

Workshop Objective: To identify and map where the community's food supply comes from, and how these channels of food access can be disrupted or strengthened by factors within and outside of the community, with particular focus directed at how the food system may be susceptible to the impacts of climate change.

Workshop Process: The steps below outline a participatory process for getting community members' insights, and contributions about their local food system.

First, the facilitators will explain that answering the questions that are asked is optional and they will emphasize that participants can ask for clarification whenever they would like. The facilitators will offer to transcribe notes to add to the map for matters of accessibility. Participants will also be asked to share their own questions or insights to enrich the discussion if they would like.

Participants will first be asked to identify where the community's food supply comes from and what it consists of, using the map to determine how food gets to the community and to families. AVAILABILITY

- Where are community food sources (market, trading, bartering, community meals, community store, food program, wild, gardened)?
- What type of foods are available and eaten by you and the community?

| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| 1 | System | Subject | Significance Statements (Grouped by colour for theme develo | Frequency | SWOT | Communities | Number of Par |
| 2 | Market | Shopping Strategies | Major cities. Thunder Bay has cheaper groceries and more sel | 14 | O/T, O- | BZA, NN, RRIB | |
| 3 | Market | Shopping Strategies | Preference for certain food retailers over others. Perceived fo | 13 | O- peop | BZA, NN, RRIB | |
| 4 | Market | Health | Lack of healthy food options nearby. | 3 | T- peop | BZA, NN | |
| 5 | Market | Shopping Strategies | Local meat/specific meat purchasing behaviours are prioritized | 8 | O- peop | BZA, NN, RRIB | |
| 6 | Market | Shopping Strategies/Transportation | Reduce number of trips to the store to save money. | 3 | W- peo | NN, BZA | |
| 7 | Market | Transportation | Reliance on ride services. | 5 | T-ride s | NN, BZA | |
| 8 | Market | Transportation | Walking is impossible/unsafe | 3 | T | RRIB, BZA | |
| 9 | Market | Transportation/Cost | Cost of transportation is prohibitive. | 8 | T | NN, BZA | |
| 10 | Market | Transportation | Transportation options are not always reliable. | 6 | T | RRIB, BZA | |
| 11 | All | Health | Poor health is linked to changes in food consumption. | 7 | W | BZA, NN, RRIB | |
| 12 | All | Health | Desire to make healthy food choices. | 4 | S | BZA | |
| 13 | All | Health | Changing food consumption behaviour is a challenge. | 3 | W | BZA, NN, RRIB | |
| 14 | All | Programs | Community services are utilized. | 17 | O | BZA, NN, RRIB | |
| 15 | All | Community | Social connections are important food resources. | 20 | S/W | BZA, NN, RRIB | |
| 16 | All | Solutions | Unconventional/New food system strategies | 7 | S | BZA, NN, RRIB | |
| 17 | All | Programs | Underfunded or understaffed community services | 4 | O/T | BZA, NN, RRIB | |
| 18 | Garden/Wild | Knowledge and Capacity | Loss of knowledge and capacity | 33 | W/T | BZA, NN, RRIB | |
| 19 | Garden | Knowledge and Capacity | Some community members have their own gardens and/or anir | 4 | S | RRIB, BZA | |
| 20 | Wild | Food System Dynamics | Food harvest locations are kept a secret, which is only shared | 2 | W | RRIB, BZA | |
| 21 | Garden/Wild | Perception of Health | Gardened food and/or food from the wild is percieved as heal | 7 | S/O/T | BZA, NN, RRIB | |
| 22 | All | Food Strategies | Preserving food to make it last. | 3 | S/O/T- | BZA | |
| 23 | Wild | Environmental Change | Changes in the environment are changing dietary patterns or a | 5 | T-chang | BZA, NN | |
| 24 | Wild | Knowledge and Capacity | People interact with their environments to increase wild food s | 1 | S- comr | BZA | |
| 25 | Market | Shopping Strategies | Bulk shopping is considered an important food security strate | 4 | | BZA, NN, RRIB | |
| 26 | Wild | Environmental Change/Capacity | Natural resource conflicts | 5 | | BZA, NN, RRIB | |
| 27 | Garden/Wild | Culture/Knowledge and Capacity | Food is a key part of people's connection to the natural world | 16 | | BZA, NN, RRIB | |
| 28 | Garden/Wild | Culture/Knowledge and Capacity | Engaging in hands on/land-based practices is central to learni | 10 | | BZA, (1)NN, RRIB | |
| 29 | Garden/Wild | Food Strategies | Focus on community self-sufficiency (sometimes in the case of | 9 | | BZA, (1)NN, RRIB | |
| 30 | Garden/Wild | Solutions | Hope is placed in the children | 7 | | BZA, NN, RRIB | |
| 31 | Wild | Environmental Change/Capacity | There are concerns about glyphosate in food or impacting foc | 5 | | BZA, NN | |
| 32 | Wild | Environmental Change/Capacity | Some wild foods are unsafe or are percieved as being unsafe. | 5 | | BZA, (1)NN | |
| 33 | All | Trust/Colonialism | Lack of trust in the government and/or industry. | 10 | | BZA, NN, (1)RRIB | |
| 34 | Wild | Environmental Change | Species abundance is changing, but it is unclear whether it is c | 4 | | BZA, NN | |

Thematic Analysis of Focus Group, Interview, and Participatory Mapping Data

# CLIMATE CHANGE AND TRADITIONAL FOODS

## WOODLAND ANIMALS AND GROUSE

### KEY POINTS

CLIMATE CHANGE WILL IMPACT THE HEALTH OF THE LAND:

- DISTURBANCES SUCH AS INSECT INVASIONS, ICE STORMS, AND FIRE MAY BECOME MORE FREQUENT
- CHANGES IN SEASONALITY: DECLINING AVERAGE WINTER SNOW DEPTH, LATER FALLS, HOTTER SUMMERS
- CHANGING SPECIES DYNAMICS: COMPETITION BETWEEN NEWLY ENCROACHING SPECIES AND EXISTING ONES
- DISRUPTED FOOD CHAINS: CHANGES IN THE LIFE CYCLES OF MANY SPECIES AND THEIR ABUNDANCE
- CHANGING PRECIPITATION PATTERNS: INSTEAD OF SNOW DURING THE WINTER, FREEZING RAIN MAY BECOME MORE COMMON. THE TIMING AND INTENSITY OF PRECIPITATION MAY ALSO CHANGE.
- WETLANDS IN THE EASTERN REGION OF THE LAKE SUPERIOR BASIN MAY BECOME DRIER

TRADITIONAL FOOD SPECIES WILL EXPERIENCE MORE STRESS AND DECREASING HABITAT SUITABILITY. UNDERSTANDING HOW THEY MAY RESPOND TO CLIMATE CHANGE IS CRITICAL TO BUILDING RESILIENCE

### WOODLAND ANIMALS

#### WHITE TAILED DEER

White-tailed deer currently live alongside moose in the region surrounding Netmizaaggamig Nishnaabeg, Biinjitiwaabik Zaaging Anishinaabek, and Red Rock Indian Band. Research by Ranta & Lankester (2017) in the Kenora District (beside the Thunder Bay district on its Northwestern edge) identified that harsh winter conditions decrease deer populations. Snow depth is one of the most important factors for deer survival, because they have difficulty moving through snow that is deeper than 40cm, with snow deeper than 75cm being life-threatening (Kennedy-Slaney et al., 2018; Ranta & Lankester, 2017). Ranta & Lankester (2017) observed that the rise in the population of deer in the Kenora District between the 1990s and 2000s was directly tied to winter conditions that favoured the health of deer for several years in a row (2017). If average winter snow depth continues to decline near Netmizaaggamig Nishnaabeg, Biinjitiwaabik Zaaging Anishinaabek, and Red Rock Indian Band, deer may become more common in the area (Kennedy-Slaney et al., 2018).

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# CLIMATE CHANGE AND TRADITIONAL FOODS

## FRESHWATER FISH

### KEY POINTS

ASIDE FROM THE VARIED IMPACTS THAT CLIMATE CHANGE WILL HAVE ON THE LAND ARE THE IMPACTS IT WILL HAVE ON THE WATER, INCLUDING:

- DECLINES IN DISSOLVED OXYGEN CONTENT: INCREASED ALGAE GROWTH, DECLINING FISH HEALTH
- WARMER WATER TEMPERATURES: LESS TIME FOR FISH TO FORAGE IN THEIR OPTIMAL CONDITIONS
- CHANGES IN SEASONALITY: CHANGING SEASON LENGTH, AVERAGE TEMPERATURE, AND ICE TIMING
- DISRUPTED FOOD CHAINS: CHANGES IN THE LIFE CYCLES OF MANY SPECIES AND THEIR ABUNDANCE

FISH WILL EXPERIENCE MORE STRESS AND DECREASING HABITAT SUITABILITY. UNDERSTANDING HOW FISH MAY RESPOND TO CLIMATE CHANGE IS CRITICAL TO BUILDING RESILIENCE

### COLD-WATER FISH

#### LAKE TROUT

Lake trout are a cold-water fish species that require relatively high dissolved oxygen content and low nutrient levels to thrive (Dove-Thompson et al., 2011). As a cold-water fish, lake trout are susceptible to climate stress (Collingsworth et al., 2017). In a study in Northwestern Ontario, by Guzzo & Blanchfield (2017), a trend of longer springs and later falls was observed over a forty-three-year time span. This change in seasonality caused a temporary loss in the best lake trout habitat, this loss becoming more common over time (Guzzo & Blanchfield, 2017). The loss of the best habitat could cause the health of lake trout to decline, particularly if hotter summers also place stress on the fish population (Guzzo & Blanchfield, 2017; Jansen & Hesslein, 2004). This is significant because lake trout spawn during the fall, which requires fish to have had enough food in spring and summer to support reproduction (Jansen & Hesslein, 2004). Potential shifts in nutrient and dissolved oxygen content, and species invasions are also driven by warming temperatures, and are placing stress on Lake trout populations (Nelligan et al., 2016; Sharma et al., 2009).

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# CLIMATE CHANGE AND TRADITIONAL FOODS

## MUSHROOMS AND BERRIES

### BERRIES

#### BLUEBERRIES

A study by Dyukaryeva & Mallik (2023) found that shaded conditions result in lower berry yield and lower antioxidant activity in two wild blueberry species in Northwestern Ontario, velvet-leaved blueberry and lowbush blueberry. A boreal forest landscape where forest fires are more frequent may create conditions that are more conducive to the growth of some blueberry plants and to their nutritional quality.

#### STRAWBERRIES

The climate sensitivity of wild strawberries near Netmizaaggamig Nishnaabeg, Biinjitiwaabik Zaaging Anishinaabek, and Red Rock Indian Band is understudied. One study by Lehmberg et al. (2022) considered how spring bloom time of wild strawberries has shifted in response to climate change. Their results indicate that wild strawberry has experienced an average 53-day lag in spring bloom time between 1949 and 1992, which might be due to the berry plants not being exposed to the necessary number of cold days throughout the winter. The researchers also found that as the number of days with temperatures greater than 0°C increased between February 1st and May 31st, strawberries flowered earlier. This inconsistency in their results is likely because of how much climate varies from year to year. It is difficult to know whether warmer winters or warmer springs will have a larger impact on wild strawberry flowering time, and perhaps wild strawberry fruit production (Lehmberg et al., 2022).

#### LOW BUSH CRANBERRIES

A study by Hirabayashi et al. (2022) modelled how three wild berry species, including small cranberry, may respond to climate change. Reassuringly, all of the species included in the study are expected to maintain stable ranges within the area surrounding Netmizaaggamig Nishnaabeg, Biinjitiwaabik Zaaging Anishinaabek, and Red Rock Indian Band (Hirabayashi et al., 2022). However, small cranberry performs the worst of the wild species because it grows in boggy areas (Hirabayashi et al., 2022). Findings from a study by Chu (2015) predict that wetlands in the eastern region of the Lake Superior basin will be vulnerable to drying, impacting small cranberry growth near Netmizaaggamig Nishnaabeg.

#### RASPBERRIES

Wild raspberries are understudied in Ontario. Slaughter & Schultz (2012) characterize arctic raspberry as vulnerable to climate change, primarily due to changes in water balance, increased risk of fires, and changes in how plants interact with each other. This does not explain how other raspberry types may adapt to climate change. At the southern edge of the boreal, arctic raspberries are limited to Sphagnum-dominated fens (a type of wetland), while other species are not (Slaughter & Schultz, 2012). However, Sphagnum-dominated fens also support other berry species, including small cranberry, creeping snowberry, and lowbush blueberry, making the findings from Slaughter & Schultz (2012) telling of how other traditional foods may respond to climate change.

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